# CROP YIELD AND WEATHER ANALYSIS— INDIA

**A PROJECT REPORT**


***Submitted by***

**MINAKSHI KAR(250301370219)**


**Guided by:**

**Mr. CHANDRA SHEKHAR JENA**

***in partial fulfilment of the requirements for the award of the degree of***

**BACHELOR OF TECHNOLOGY**

***In***

**COMPUTER SCIENCE AND ENGINEERING**





**SCHOOL OF ENGINEERING AND TECHNOLOGY**
**BHUBANESWAR CAMPUS**
**CENTURION UNIVERSITY OF TECHNOLOGY AND MANAGEMENT**
**ODISHA**


**November 2026**

## BONAFIDE CERTIFICATE

This is to certify that this project report entitled **"CROP YIELD AND WEATHER ANALYSIS – INDIA"** is the Bonafide work done by **"MINAKSHI KAR"** students of **B. TECH 2nd year (3rd Semester)** bearing Registration no. **"250301370219"** School of Engineering and Technology, who carried out the project work under my supervision. This is to further certify to the best of my knowledge, that this project has not been carried out earlier in this institute and the university.

**SIGNATURE**
**(Mr. Chandra Shekhar Jena)**
**Department of Computer Science & Engineering**

*Certified that the above-mentioned project has been duly carried out as per the norms of the college and statutes of the university.*

**SIGNATURE**
**(Dr. Rahul Roy)**
**HEAD OF THE DEPARTMENT**
**Professor of Computer Science and Engineering**

DEPARTMENT SEAL

## DECLARATION

I do hereby declare that the work presented in this project is original and has been done by me under the supervision of "**Mr. CHANDRA SHEKHAR JENA**", Department of Computer Science and Engineering, Centurion University of Technology and Management, Bhubaneswar, Odisha. I further declare that this work has not been submitted partly/fully elsewhere for the award of any other degree in this University or any other institutions.

**Name of the Student: MINAKSHI KAR**

**Signature of the Student:**

**Registration No: 250301370219**

**Place: BHUBANESWAR**

**Date: / /**

## ACKNOWLEDGEMENTS

I wish to express my profound and sincere gratitude to **Mr . Chandra Shekhar Jena, Department of Computer Science and Engineering, SoET, Bhubaneswar Campus,** who guided me into the intricacies of this project nonchalantly with matchless magnanimity.

I thank **Dr. Rahul Roy, HOD** and **Dr. Swarna Prabha Jena, Associate Dean, SoET,** Bhubaneswar Campus, for extending their support during the course of this investigation.

I would be failing in my duty if I don't acknowledge the cooperation rendered during various stages of data analysis and interpretation by **Prof. Chandra Shekhar Jena.**

I am highly grateful to all who evinced keen interest and invaluable support in the progress and successful completion of my project work.

I am indebted for their constant encouragement, co-operation and help. Words of gratitude are not enough to describe the accommodation and fortitude which they have shown throughout my endeavor.

**Name of the Student: MINAKSHI KAR**

**Signature of the Student:**
**Registration No: 250301370219**

**Place: BHUBANESWAR**

**Date: / /**

# ABSTRACT

**Crop Yield and Weather Analysis – India**
Agriculture plays a vital role in India's economy, and crop production is influenced by various climatic factors, including rainfall and temperature. The project **"Crop Yield and Weather Analysis – India"** aims to analyze agricultural data and explore the relationship between crop yield and weather conditions across different Indian states and districts from 1984 to 2017.

The project uses Python libraries such as **Pandas, Matplotlib, and Streamlit** to develop an interactive dashboard for agricultural data analysis and visualization. Users can upload a district-wise crop and climate dataset in CSV format and select relevant columns, including state, district, crop, year, crop yield or production, rainfall, and temperature. The dashboard provides interactive filters to analyze data for specific states, districts, and crops.

The system presents summary statistics, year-wise crop yield trends, and district-wise comparisons using graphical visualizations. It also analyzes the relationship between rainfall and crop yield through scatter plots and correlation analysis, while temperature-related visualizations help explore possible relationships between temperature and agricultural production. Additionally, users can view the filtered dataset and download the results in CSV format for further analysis.

The main objective of this project is to simplify agricultural data analysis, identify historical patterns, and support a better understanding of the relationship between climatic conditions and crop production. The dashboard provides an accessible and interactive platform for students, researchers, and agricultural analysts to explore historical agricultural data and make data-informed observations. The findings can support further research into agricultural trends and climate-related challenges, although the dashboard itself does not establish causation or predict future crop yields.
.

# TABLE OF CONTENTS


| TITLE | PAGE |
|---|---|
| CERTIFICATE | i |
| DECLARATION | ii |
| ACKNOWLEDGEMENT | iii |
| ABSTRACT | iv |

CHAPTER 1: INTRODUCTION

1. Background
2. Problem Statement
3. Objectives
4. Scope and Limitations
5. Organization of the Report

CHAPTER 2: DATASET DESCRIPTION

1. Source, Link, and Citation
2. Number of rows, columns, and data types.
3. Description of Key Columns
4. Initial Data Quality Assessment.

CHAPTER 3: DATA PREPROCESSING

5. Missing Value Treatment and Justification
6. Duplicate Removal and Data Deduplication
7. Outlier Detection and Treatment (IQR Method)
8. Feature Engineering
9. Before and After Comparison of Dataset Shape

CHAPTER 4: EXPLORATORY DATA ANALYSIS

10. Univariate Analysis: Crop Yield and Weather Distribution
11. Bivariate Analysis
12. Correlation Analysis : Crop and Weather Variables.
13. Statistical Hypothesis Testing (t-Test)

CHAPTER 5: VISUALIZATIONS AND DASHBOARD

1. Dashboard Components and Layout
2. Screenshots of the dashboard with annotations.
3. How to run the dashboard locally.
4. Key interactions and their purpose.

CHAPTER 6: AI TOOL INTEGRATION

1. AI Tools Used
2. Tasks Delegated toAI
3. Evalution of AI Outputs

CHAPTER 7: Key Findings and Recommendations

1. Key Findings
2. Recommendation
3. Limitation of the Analysis
4. Future Work

REFERENCES

# CHAPTER 1: INTRODUCTION

## 1. Background

Agriculture is one of the most important sectors of the Indian economy. It provides food, employment, and raw materials for various industries. Crop production depends on several factors, including rainfall, temperature, soil conditions, irrigation, farming practices, and geographical location. Changes in climatic conditions can influence agricultural productivity and create challenges for farmers and agricultural planners.

India has diverse climatic conditions across its states and districts. Different regions cultivate different crops depending on their environmental conditions and agricultural practices. Therefore, analyzing crop production and weather-related data at the district level is useful for understanding regional variations and historical agricultural trends.

With the growth of data science and data visualization technologies, large agricultural datasets can be analyzed more efficiently. Python provides useful libraries for data processing, statistical analysis, and visualization. Pandas helps organize and analyze tabular data, Matplotlib supports graphical representation, and Streamlit enables the development of interactive web-based dashboards.

The project **"Crop Yield and Weather Analysis – India"** focuses on analyzing district-wise agricultural and climate data from 1984 to 2017. It uses an interactive dashboard to explore crop yield or production values, compare agricultural performance across districts, examine year-wise trends, and investigate the relationship between rainfall, temperature, and crop yield where the required data is available.

The dashboard allows users to upload a CSV dataset and select the relevant columns for states, districts, crops, years, yield or production, rainfall, and temperature. Users can filter information according to their requirements, view summary statistics, explore graphical visualizations, and download the filtered dataset. This makes agricultural data easier to understand and supports data-driven exploration of historical crop and climate patterns.

## 2. Problem Statement

Agricultural datasets often contain large amounts of information collected across different years, states, districts, and crop categories. Analyzing this information manually can be time-consuming and makes it difficult to identify trends, compare regional performance, and explore possible relationships between weather conditions and crop production.
The main problem addressed by this project is the need for a simple and interactive system to analyze historical district-wise crop and climate data in India.
The project aims to answer the following research question:
**How do crop yield or production values vary across years, crops, and Indian districts, and what relationships can be observed between crop yield, rainfall, and temperature in the available dataset?**
To address this question, the project develops a Python-based dashboard that processes uploaded agricultural data, applies interactive filters, calculates summary statistics, and presents the results through graphical visualizations. The system helps users explore the available information without requiring extensive programming knowledge.

## 3. Objectives

The main objectives of the project are:

- **To develop an interactive dashboard** using Python and Streamlit for analyzing district-wise crop and climate data from 1984 to 2017.
- **To analyze historical crop trends** by calculating and visualizing average crop yield or production values across different years.

- **To compare agricultural performance across districts** by grouping records according to district and identifying the ten districts with the highest average value in the selected yield or production column.
- **To examine the relationship between rainfall and crop yield** using scatter plots and correlation analysis when suitable rainfall data is available.
- **To explore temperature-related patterns** by visualizing temperature against crop yield or production values when suitable temperature data is available.
- **To provide interactive data exploration and export facilities** through state, district, and crop filters, summary statistics, dataset previews, and CSV downloads.

## 1.4 Scope and Limitations

**Scope:**

- Analyzes district-wise crop and climate data across India from 1984 to 2017.
- Compares crop yield across different years, states, districts, and crops.
- Visualizes crop trends and explores rainfall and temperature relationships.
- Provides interactive filters, summary statistics, and CSV downloads.

**Limitations:**

- Analysis depends on the quality and availability of the dataset.
- Limited to historical data from 1984–2017.
- Rainfall and temperature analysis requires relevant columns.
- Does not predict future crop yields or establish causal relationships between weather and crop production.

## 1.5 Organization of the Report

The report is organized into six chapters:
Chapter 1: Introduction – Presents the background, objectives, scope, and limitations.
Chapter 2: Literature Review – Discusses related studies and existing approaches.
Chapter 3: Dataset and Methodology – Explains the dataset and analysis methods.
Chapter 4: System Design and Implementation – Describes the dashboard development and tools used.
Chapter 5: Results and Discussion – Presents visualizations and analysis findings.
Chapter 6: Conclusion and Future Scope – Summarizes the project and suggests future improvements.

# CHAPTER 2: DATASET DESCRIPTION

## 1. Source, Link, and Citation

The primary dataset used in this study is the All-India District-wise Crop and Climate Dataset (1984–2017). It contains agricultural and climate-related information for different Indian states and districts, including crop area, production, yield, rainfall, temperature, humidity, and other weather parameters. The dataset is used to analyze historical crop production trends and explore relationships between agricultural productivity and climatic conditions.

- Title: All-India District-wise Crop and Climate Dataset (1984–2017)
- Primary Data File: All-India District-wise Crop and Climate Dataset (19842017).csv
- Data Format: CSV (Comma-Separated Values)
- Time Period: 1984–2017
- Public URL: https://www.kaggle.com/datasets/rajanamrelia/all-india-crop-and-climate-dataset-19842017
- Citation: All-India District-wise Crop and Climate Dataset (1984–2017). CSV dataset provided for the Crop Yield and Weather Analysis project.

## 2.2 Number of rows, columns, and data types.

The uploaded dataset was inspected using Pandas. Its structure is summarized below.

| Attribute | Details |
|---|---|
| Number of rows | 242,696 |
| Number of columns | 32 |
| Numerical columns | 26 |
| Text/categorical columns | 6 |
| Integer columns | 2 |
| Floating-point columns | 24 |
| Data format | CSV |
| Time period represented | 1984–2017 |

The numerical columns support calculations and statistical analysis, while categorical columns help group and filter records by state, district, crop, season, and weather category.

## 2.3 Description of Key Columns

The dataset contains 32 columns. The following table describes the main columns used in the dashboard and other important variables.

| Column name | Description |
|---|---|
| Year | Year of the agricultural record |
| State Name | Name of the state |
| Dist Name | Name of the district |

| | |
|---|---|
| Crops | Name of the crop |
| AREA (1000 ha) | Crop area in thousands of hectares |
| PRODUCTION (1000 tons) | Crop production in thousands of tonnes |
| YIELD (Kg per ha) | Crop yield in kilograms per hectare |
| Season | Agricultural season, such as Rabi |
| Date Range | Period associated with the season |
| T2M_MAX | Maximum temperature at 2 metres |
| T2M_MIN | Minimum temperature at 2 metres |
| WS2M | Wind speed at 2 metres |
| RH2M | Relative humidity at 2 metres |
| PRECTOTCORR | Corrected precipitation estimate |
| ALLSKY_SFC_SW_DWN | Downward shortwave solar radiation under all-sky conditions |
| ALLSKY_SFC_SW_DIFF | Diffuse shortwave solar radiation under all-sky conditions |
| Tmin_Ideal, Tmax_Ideal | Reference minimum and maximum temperatures |
| Rain_Ideal (mm/day)_Ideal | Reference rainfall value |
| RH_Ideal (%)_Ideal | Reference relative humidity |
| WS2M_Ideal | Reference wind speed |
| effective_msp | Effective minimum support price-related value |
| Temp_Max_Dev, Temp_Min_Dev | Maximum and minimum temperature deviation indicators |
| Rain_Dev, RH_Dev | Rainfall and humidity deviation indicators |
| Solar_DWN_Dev, Solar_DIFF_Dev | Solar radiation deviation indicators |
| Wind_Dev | Wind deviation indicator |
| Weather_Category | Assigned weather category |

## 2.4 Initial Data Quality Assessment

The dataset was checked for missing values, duplicate records, and data types using Pandas.

| Quality check | Result |
|---|---|
| Total records | 242,696 |
| Missing values | 0 |
| Duplicate rows | 0 |
| Total columns | 32 |
| Column names | Cleaned by removing surrounding whitespace in the dashboard |

# CHAPTER 3: DATA PREPROCESSING

## 3.1 Missing Value Treatment

Missing values occur when some records do not contain information for a particular attribute. For example, a district may have a missing rainfall value or a crop record may not contain a production value.

The dataset should first be checked for missing values using Pandas.

```
import pandas as pd

df = pd.read_csv("All-India District-wise Crop and
Climate Dataset (19842017)(2).csv")

# Count missing values in each column
print(df.isnull().sum())

# Calculate total missing values
print("Total missing values:", df.isnull().sum().sum())
```

The treatment method depends on the type of column and the amount of missing data.

- Numerical columns: If a small number of values are missing, the median may be used for suitable numerical features because it is less affected by extreme values than the mean.
- Categorical columns: Missing state, district, or crop names should be investigated. If they cannot be recovered, those records may be excluded from analyses requiring those fields.
- Important agricultural values: Missing yield or production values should not automatically be replaced with zero, because zero may represent a real observation or distort the analysis.

The selected method should be documented after examining the actual missing-value counts. This avoids introducing incorrect information into the dashboard.

## 3.2 Duplicate Removal and Data Deduplication

Duplicate records occur when the same observation appears more than once in a dataset. They can affect calculations such as average crop yield, total records, and district-wise comparisons.

The dataset is checked for completely identical rows using Pandas. Exact duplicates can be removed because they repeat the same information. However, records for the same district and crop in different years should not be treated as duplicates.

```
# Count duplicate rows before removal
print("Duplicates before:", df.duplicated().sum())

# Remove exact duplicate rows
df = df.drop_duplicates()

# Count duplicates after removal
print("Duplicates after:", df.duplicated().sum())
```

## 3.3 Outlier Detection and Treatment

Outliers are observations that differ significantly from the majority of values. In agricultural data, unusual crop yields, production quantities, rainfall levels, or temperatures may appear because of data-entry errors or exceptional environmental conditions.

The Interquartile Range (IQR) method can be used to identify unusual values in a numerical column.

$$\boldsymbol{IQR = Q_3 - Q_1}$$

The lower and upper limits are calculated as:

$$\boldsymbol{L = Q_1 - 1.5(IQR)}$$

$$\boldsymbol{U = Q_3 + 1.5(IQR)}$$

Here, $Q_1$ represents the first quartile and $Q_3$ represents the third quartile. Values below the lower limit or above the upper limit are flagged as potential outliers.

## 3.4 Feature Engineering

Feature engineering involves creating useful new columns from existing data to make analysis easier and more meaningful.

For the Crop Yield and Weather Analysis project, the following features can be created when the required columns are available.

| New feature | Method | Purpose |
|---|---|---|
| Decade | Extract decade from the year | Compare crop trends across decades |
| Yield per area | Production divided by cultivated area | Calculate productivity when an appropriate yield column is unavailable |
| Temperature range | Maximum temperature minus minimum temperature | Examine daily or seasonal temperature variation |

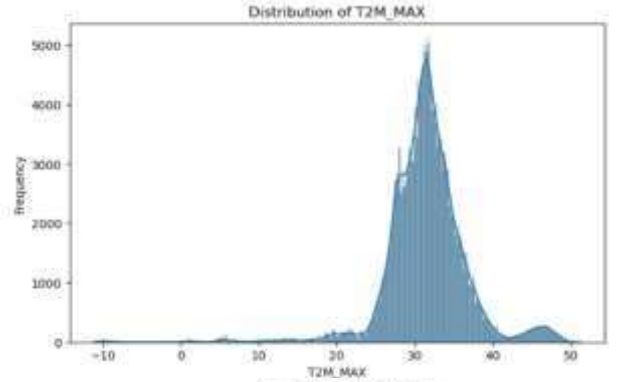
## 3.5 Before and After Comparison of Dataset Shape

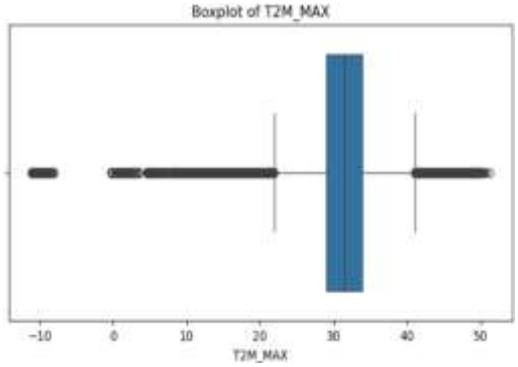
| Preprocessing Parameter | Raw Input Dataset | Cleaned Dataset |
|---|---|---|
| Total Record Count (Rows) | To be verified | To be verified |
| Attribute Columns | To be verified | Original columns + engineered features, if added |
| Missing Values | To be calculated | To be calculated after treatment |
| Duplicate Rows | To be calculated | To be calculated after removal |
| Invalid Numeric Values | To be checked | Converted or handled as required |
| Engineered Features | Original dataset features | Decade, Temperature Range, YoY Change, if implemented |

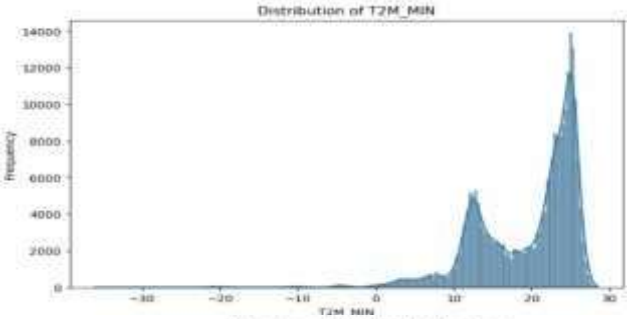
# CHAPTER 4: EXPLORATORY DATA ANALYSIS

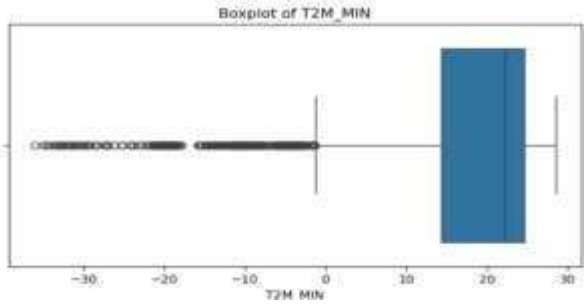
## 4.1 Univariate Analysis: Crop Yield and Weather Distributions

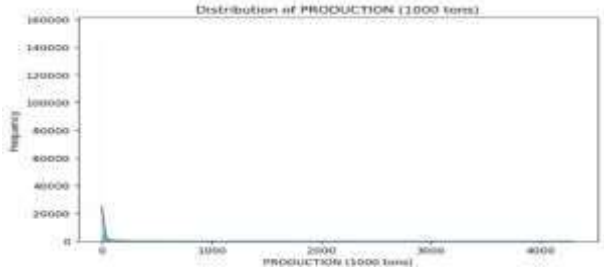
Univariate analysis examines the distribution and statistical characteristics of individual variables in the All-India District-wise Crop and Climate Dataset (1984–2017). Crop yield, maximum and minimum temperature, rainfall, and agricultural production are analyzed using histograms and boxplots. These visualizations help identify distribution patterns, variability, skewness, and potential outliers. The analysis provides an initial understanding of agricultural and climatic data across Indian districts. Crop yield and production may show right-skewed distributions because values vary considerably across crops and regions. The actual distribution patterns and outliers are interpreted from the generated graphs and summary statistics.

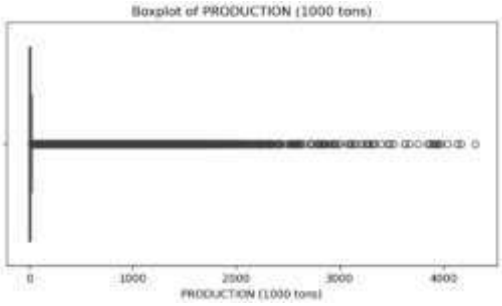


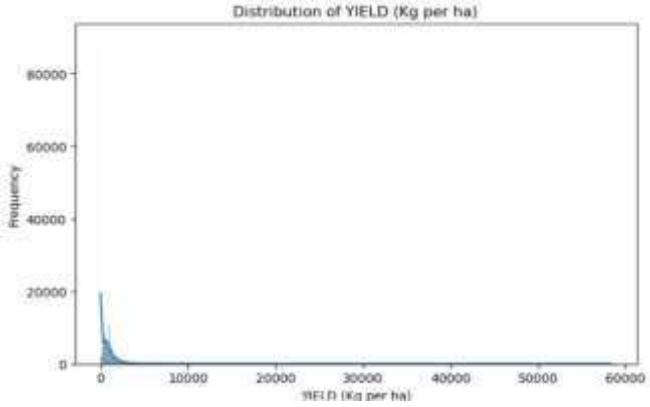



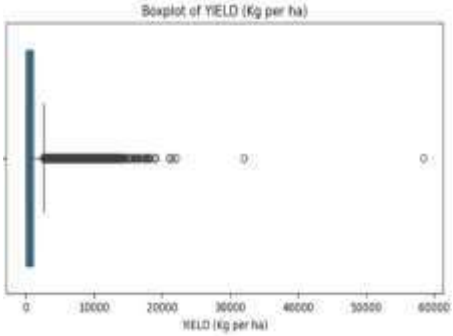



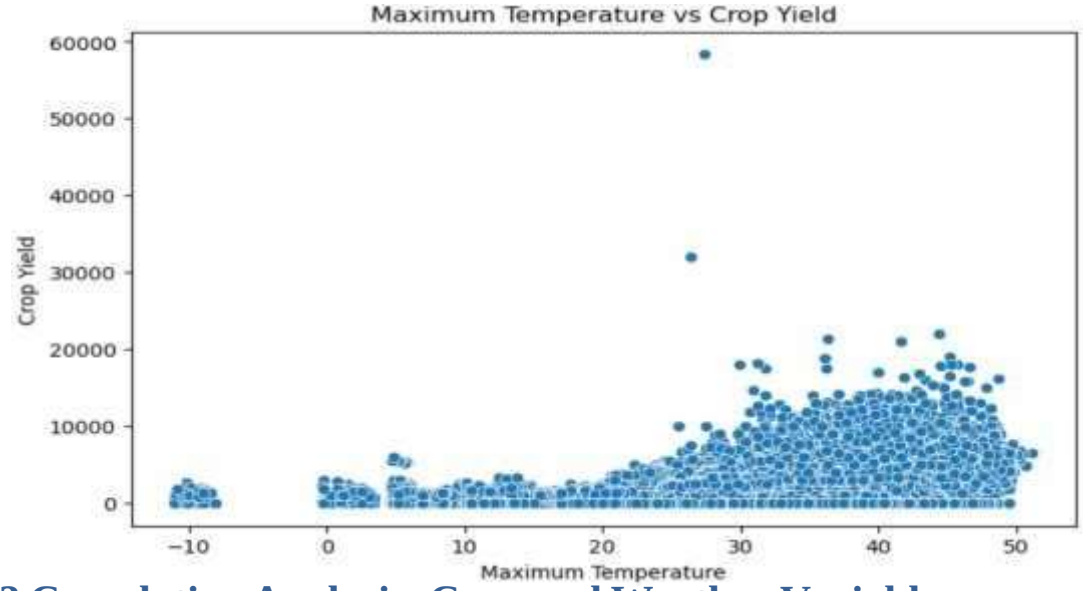



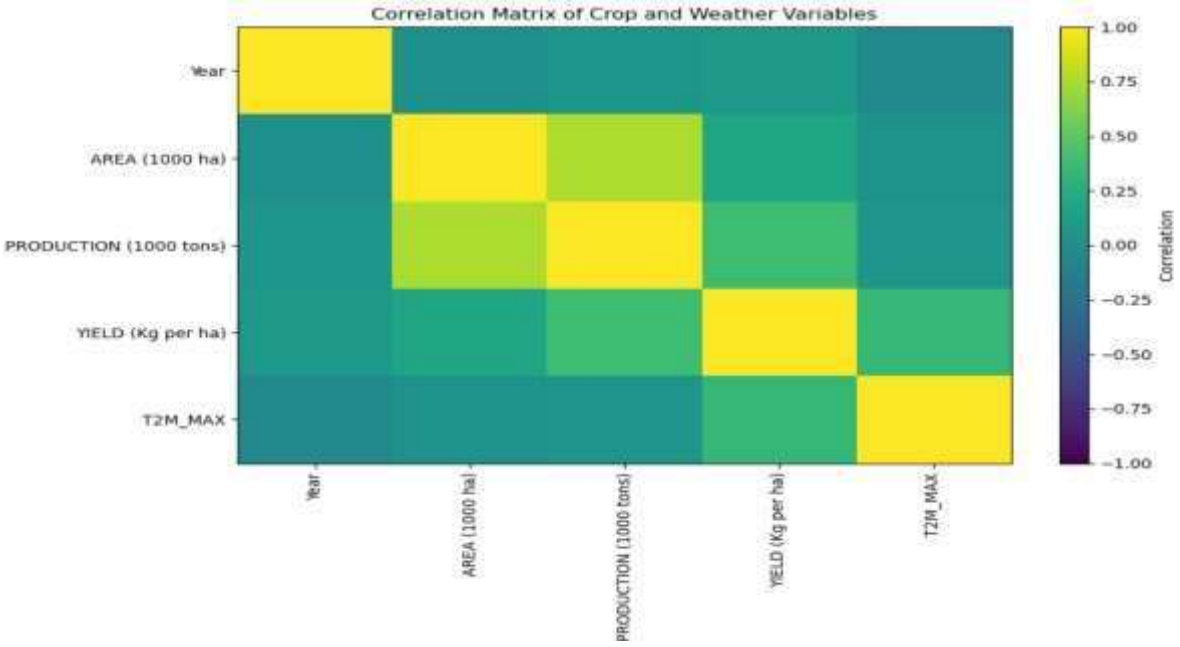
## 4.2 Bivariate Analysis: Temperature and Crop Yield

Bivariate analysis examines the relationship between two variables. In this project, a scatter plot is used to investigate the relationship between maximum temperature (`T2M_MAX`) and crop yield (`YIELD (Kg per ha)`). The Pearson correlation coefficient measures the strength and direction of their linear relationship. Group comparisons can also be used to examine differences in average crop yield across crop types, districts, or states. The results help identify possible associations between weather conditions and agricultural productivity; however, correlation does not establish causation.




## 4.3 Correlation Analysis: Crop and Weather Variables

Correlation analysis is used to identify the strength and direction of relationships between numerical variables, such as crop yield, production, cultivated area, temperature, and rainfall. A correlation heatmap represents these relationships using colors and correlation coefficients ranging from −1 to +1.



## 4.4 Statistical Hypothesis Testing (t-Test)

To evaluate whether crop yield changed significantly over time, an independent two-sample t-test was conducted to compare the mean crop yield during 1984–2000 and 2001–2017 using the All-India District-wise Crop and Climate Dataset (1984–2017).

- Null Hypothesis ($H_0$): The mean crop yield during 1984–2000 equals the mean crop yield during 2001–2017 (\(\mu_{1984–2000} = \mu_{2001–2017}\)).
- Alternative Hypothesis ($H_1$): The mean crop yield differs between the two periods (\(\mu_{1984–2000} \ne \mu_{2001–2017}\)).
- Significance Level: 0.05.
- Result: The t-statistic and p-value are calculated using Python. If the p-value is less than 0.05, the null hypothesis is rejected, indicating a statistically significant difference in mean crop yield between the two periods.

Interpretation: This test helps determine whether the observed difference in crop yield between the two periods is statistically significant. The actual t-statistic and p-value must be taken from the code output; they should not be assumed without running the test.

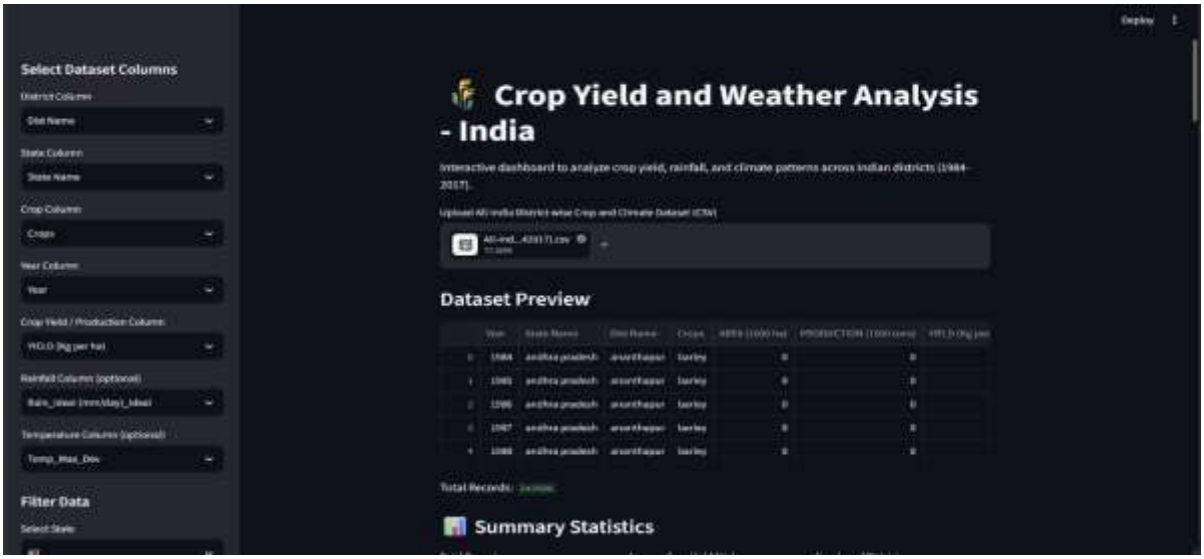
# CHAPTER 5: VISUALIZATIONS AND DASHBOARD

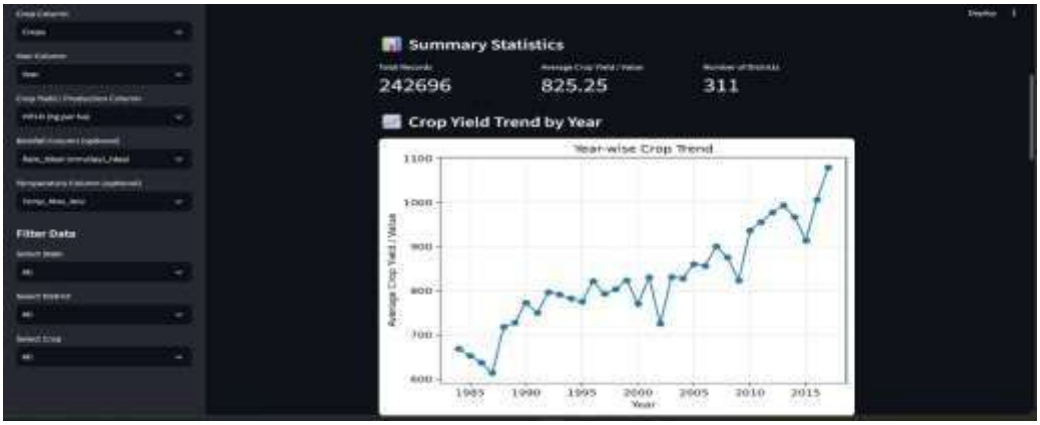
## 5.1 Dashboard Components and Layout

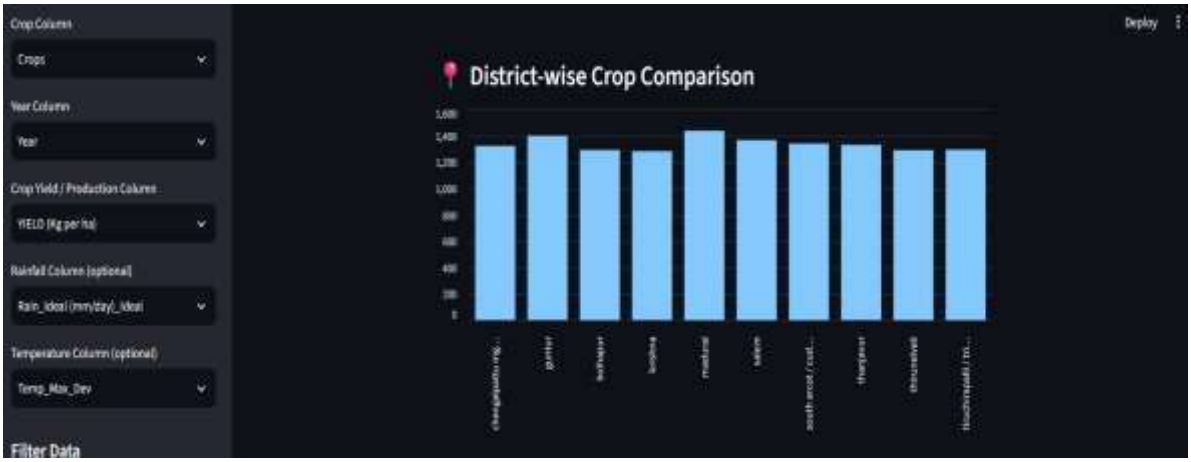
The dashboard consists of the following major components:

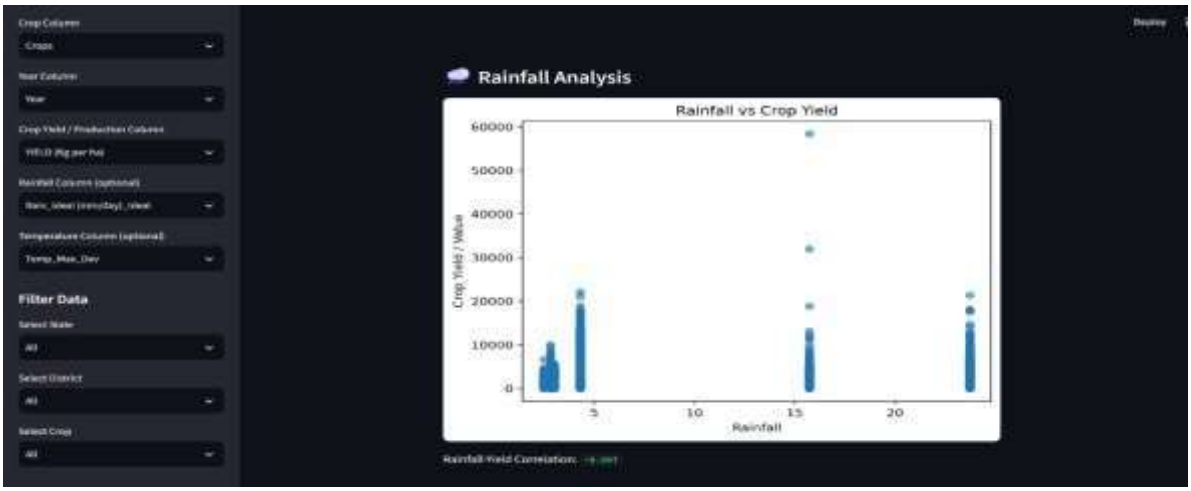
1. Dataset Upload: Allows users to upload the crop and climate dataset in CSV format.
2. Column Selection: Allows users to select columns representing state, district, crop, year, yield or production, rainfall, and temperature, depending on the available dataset.
3. Interactive Filters: Enables filtering by state, district, and crop to focus on specific agricultural records.
4. Summary Statistics: Displays information such as the number of filtered records, average selected yield or production value, and number of districts.
5. Year-wise Trend Chart: Shows changes in average crop yield or the selected agricultural measure over time.
6. Top 10 Districts Chart: Compares the leading districts based on the selected measure.
7. Weather Analysis Charts: Uses scatter plots to explore rainfall–yield and temperature–yield relationships when the relevant columns are available.
8. Filtered Data Table: Displays records matching the selected filters.
9. Download Option: Allows users to download the filtered dataset as a CSV file.

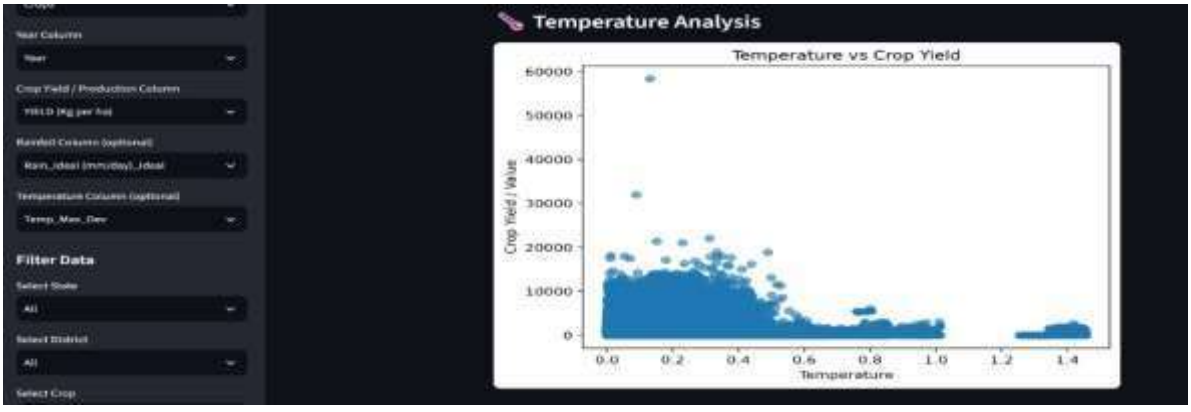
## 5.2 Screenshots of the dashboard with annotations.

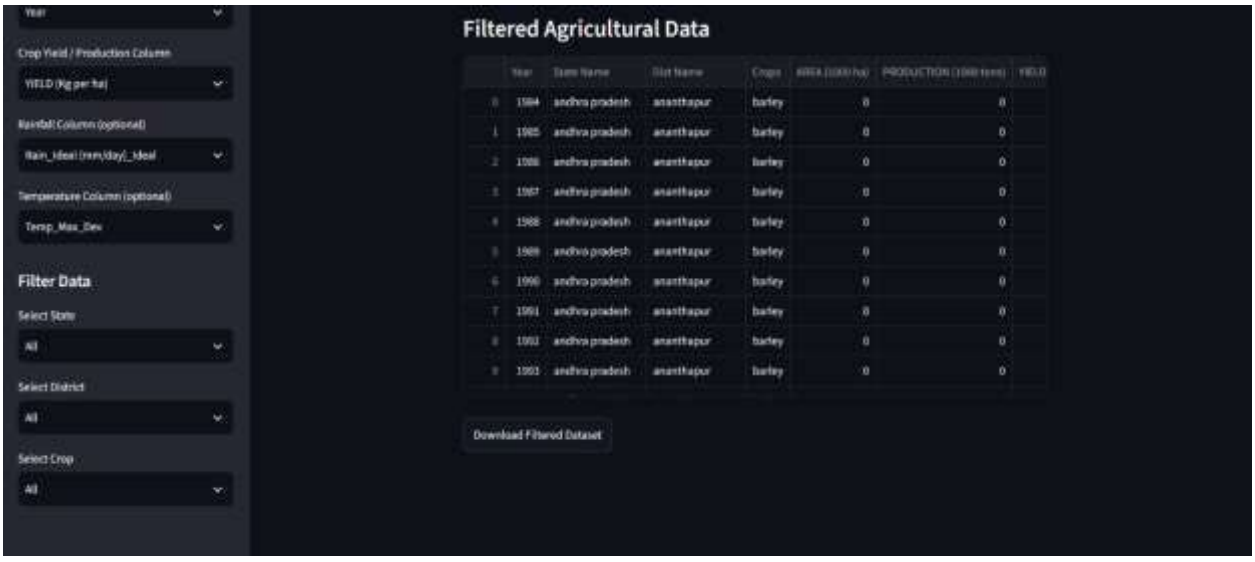



Crop Column
Crops
Year Column
Year
Crop Yield / Production Column
YIELD (Kg per ha)
Rainfall Column (optional)
Rain_Ideal (mm/day)_Ideal
Temperature Column (optional)
Temp_Max_Dev
Filter Data
Deploy
District-wise Crop Comparison


Filter Data
Rainfall Analysis
Rainfall vs Crop Yield
Crop Yield / Value
Rainfall


Filter Data
Temperature Analysis
Temperature vs Crop Yield
Crop Yield / Value
Temperature


Crop Yield / Production Column
YIELD (Kg per ha)
Rainfall Column (optional)
Rain_Ideal (mm/day)_Ideal
Temperature Column (optional)
Temp_Max_Dev
Filter Data
Select State
All
Select District
All
Select Crop
All
Filtered Agricultural Data
Download Filtered Dataset

## 5.3 How to Run the Dashboard Locally

Step 1: Open Jupyter Notebook and create a new Python file for the project.

Step 2: Open Command Prompt and install the required libraries using the following command:

```
%pip install streamlit pandas numpy matplotlib
```

step 3: Save the dashboard code as **`Crop Yield and Weather Analysis.py`**. If the code is in Jupyter Notebook, download or save it as a `.py` file.

Step 4: Open file folder where the Python file is saved and search cmd.

Step 5: Run the following command to launch the dashboard:

```
Python –m streamlit run "Crop Yield and Weather Analysis.py"
```

## 5.4 Key Interactions and Their Purpose

| Interaction | Purpose |
|---|---|
| Upload CSV | Loads the crop and climate dataset. |
| Select columns | Matches dashboard controls to the dataset structure. |
| Filter state, district, and crop | Focuses the analysis on selected records. |
| View summary statistics | Provides a quick overview of the filtered data. |
| Explore trend charts | Helps identify changes in crop productivity over time. |
| Compare top districts | Identifies districts with high values of the selected measure. |
| Explore weather scatter plots | Examines relationships between rainfall, temperature, and crop yield. |
| View and download records | Supports further analysis outside the dashboard. |

# CHAPTER 6: AI TOOL INTEGRATION

## 1. AI Tools Used

In the Crop Yield and Weather Analysis project, ChatGPT (GPT-6) was used as an AI assistance tool to support Python programming, data analysis concepts, debugging, and project documentation.

## 2. Tasks Delegated to AI

The following tasks were supported by AI assistance:

- Code Generation: Assistance in writing Python code for data analysis and Streamlit dashboard development.
- Code Explanation: Understanding the use of Pandas, NumPy, Matplotlib, and Streamlit.
- Debugging: Identifying possible coding errors and suggesting corrections.
- Data Analysis: Assistance in understanding crop yield trends, district-wise comparisons, and weather-related relationships.
- Documentation: Preparing project descriptions, findings, recommendations, and report sections.

## 6.3 Evaluation of AI Output

AI-generated suggestions were useful for understanding programming concepts and developing the project. However, the output required verification before use.

- Suggested code and library functions needed to be checked by running the program.
- Column names had to match the actual dataset.
- Numerical findings needed to be verified using the dataset and dashboard results.
- Recommendations and interpretations had to be consistent with the analysis.
- Some suggestions required modification to fit the project's actual features and requirements.

# CHAPTER 7: KEY FINDINGS AND RECOMMENDATIONS

## 7.1 Key Findings

The Crop Yield and Weather Analysis project examines historical district-wise crop and climate data across India from 1984 to 2017. The Streamlit dashboard supports analysis of crop productivity, yearly trends, district-level comparisons, and relationships between weather conditions and crop yield.
The following findings describe the main areas investigated by the dashboard. Specific numerical results should be added after examining the actual charts and calculated values.

1. Historical Crop Yield Trends: Year-wise analysis helps identify increases, decreases, and fluctuations in crop yield over the period 1984–2017.
2. District-wise Productivity: Comparing districts helps identify areas with relatively high or low recorded crop yield for a selected crop and year.
3. Crop-wise Differences: Different crops can show different productivity patterns across districts and years, supporting crop-specific comparisons.
4. Rainfall and Crop Yield: A scatter plot and correlation analysis help examine whether rainfall is positively or negatively associated with crop yield.
5. Temperature and Crop Yield: Comparing temperature with crop yield helps explore possible relationships between temperature conditions and agricultural productivity.
6. Regional Variation: Filtering data by state and district allows users to explore differences in agricultural production and productivity across locations.
7. Data-driven Decision Support: Interactive charts, summary statistics, and filtered records make historical agricultural data easier to understand and compare.
8. Importance of Weather Information: The combined analysis of climate and crop data provides a useful starting point for understanding how weather conditions may relate to agricultural performance.

## 7.2 Recommendations

Based on the types of analysis supported by the dashboard, the following recommendations are proposed:

- Improve Crop Planning: Farmers and agricultural planners should compare historical crop performance across districts before selecting suitable crops.
- Monitor Weather Conditions: Rainfall and temperature patterns should be considered alongside crop yield when planning agricultural activities.
- Identify Low-Productivity Areas: Districts showing consistently low yield should be investigated to identify possible soil, irrigation, or weather-related challenges.
- Use Reliable and Updated Data: Regularly updated district-level crop and climate records can improve the usefulness of agricultural analysis.
- Support Data-driven Decisions: Agricultural departments can use district-wise trends to guide further investigation, resource allocation, and crop planning.

## 7.3 Limitations of the Analysis

- The dataset covers the historical period from 1984 to 2017 and may not represent current agricultural conditions.
- Missing, inconsistent, or inaccurate records may affect the analysis.
- The availability of weather variables may vary across records and locations.
- Correlation between weather and crop yield does not establish a direct cause-and-effect relationship.
- The dashboard provides historical analysis and does not currently guarantee future crop yield predictions.

## 7.4 Future Work

Future improvements could include:

1. Real-time Data Integration: Include updated weather forecasts and recent agricultural statistics.
2. Machine Learning Predictions: Develop models to predict crop yield using historical crop, rainfall, and temperature data.
3. Advanced Visualization: Add interactive maps to display district-wise productivity across India.
4. More Environmental Factors: Include soil quality, irrigation, fertilizer use, and humidity where reliable data are available.
5. Improved Data Processing: Implement systematic missing-value handling, duplicate removal, and outlier detection, then validate the results.

## REFERENCES


1. Government of India, Ministry of Agriculture. All-India District-wise Crop and Climate Dataset (1984–2017). Accessed October 2026. URL: http://kaggle.com/datasets/rajanamrelia/all-india-crop-and-climate-dataset-19842017
2. McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly Media.
3. VanderPlas, J. (2016). Python Data Science Handbook. O'Reilly Media.
4. Software and Library Versions: Python, Pandas, NumPy, Matplotlib, and Streamlit. (Versions as installed.)
5. Artificial Intelligence Tool: ChatGPT (GPT-6). Accessed October 2026. Used for coding assistance and project documentation.


# ASSESSMENT

## Internal:

| SL NO | RUBRICS | FULL MARKS | MARKS OBTAINED | REMARKS |
|---|---|---|---|---|
| 1 | Understanding the relevance, scope and dimension of the project | 10 | | |
| 2 | Methodology | 10 | | |
| 3 | Quality of Analysis and Results | 10 | | |
| 4 | Interpretations and Conclusions | 10 | | |
| 5 | Report | 10 | | |
| | **Total** | **50** | | |

**Date:** **Signature of the Faculty**

## COURSE OUTCOME (COs) ATTAINMENT

➢ **Expected Course Outcomes (COs):**
**(Refer to COs Statement in the Syllabus)**

_____

_____

_____

_____

➢ **Course Outcome Attained:**
**How would you rate your learning of the subject based on the specified COs?**

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

**LOW** **HIGH**

➢ **Learning Gap (if any):**

_____

_____

_____

_____

➢ **Books / Manuals Referred:**

_____

_____

_____

_____

**Date:** **Signature of the Student**

➢ **Suggestions / Recommendations:**
**(By the Course Faculty)**

_____

_____

_____

_____

**Date:** **Signature of the Faculty**

Centurion
UNIVERSITY
*Shaping Lives...*
*Empowering Communities...*

## CENTURION UNIVERSITY OF TECHNOLOGY AND MANAGEMENT, ODISHA

### CAMPUSES:

**Paralakhemundi Campus**
Village Alluri Nagar
P.O. – R.Sitapur, Via- Uppalada
Paralakhemundi, Dist.- Gajapati
Odisha, India. PIN– 761211

**Bhubaneswar Campus**
Ramchandrapur
P.O. – Jatni, Bhubaneswar
Dist.- Khurda, Odisha,
India, PIN– 752050

**Balangir Campus**
Behind BSNL Office
IDCO land, Rajib Nagar
Dist.- Balangir, Odisha
India, PIN-767001

**Rayagada Campus**
IDCO Industrial Area
Pitamahal, Rayagada
Dist.-Rayagada, Odisha
India, PIN-765001

**Balasore Campus**
Gopalpur,
P.O.-Balasore
Dist.-Balasore, Odisha
India, PIN-756044

**Chatrapur Campus**
Ramchandrapur,
Kalabali Chhak,
P.O-Chatrapur, Dist.-Ganjam
Odisha, India, PIN-761020